Task 2 :
Common failure points of a servo motor -

- **Noisy Signal causing faults and Jitter**

After voltage step down, or long routings, or even EMI noises caused due to other high power devices in the system; the servo motor behaves abnormally. It causes jitter, unstable movements, and unresponsiveness.

- **Feedback Sensor Failure**

A servo motor is made up of a normal DC motor usually attached to a gearbox being controlled with a small onboard MCU with a potentiometer/encoder. After prolonged usage or fluctuating power, the sensor modules can get damaged.

- **Power Supply Issues**

Servos need a stable power supply. In many use cases, the input power or the supply IC reaches overcurrent states. Sometimes due to other high power electronics suddenly have current spikes causes the power rail supplying power to the servo to dip in supply causes sudden stops and jitter.

- **Stall Conditions / Overloading**

In cases of dynamic load, such as in case of a cooking robot, if the weight of the container to be moved is inducing more torque the servo is designed for - the servo reaches the stall condition causing no further movement, current spikes and resulting in burnout.

- **Grounding Issues**

The ground of the servo must be shorted to the MCU ground or the logic ground. The servo's PWM and Power Ground need a reference to calculate the position of movement based on the command given to it. A floating ground, or a different ground causes a servo to move to random positions.

Solutions to the above mentioned problems -

1. **System Noise Issue**

- **Decoupling Capacitors**

Add bulk capacitors (470–1000 μF) near servo power input. This will clean the power reducing ripple. We can also introduce LC filtering if needed.

- **Separate Zones for power and signal**

We must use separate paths for power and signal. The traces handling the servo signals must be far from the power distribution or buck conversion traces and regions in a PCB, this reduces noise due to EMI.

- **Trace width and length**

We can use thick power traces to power the servo and keep the signal lines as short as possible to reduce loss.

2. **Feedback Sensor Failure**

- **Add external feedback.**

Even if the servo already has a feedback loop. We can attach an external one, especially if we use a magnetic encoder like the AS5600. We can run a constant check loop measuring the actual movement to the sent signal. If error is seen we can either recalibrate or replace.

- **Mechanical Improvements**

Due to shaft misalignment caused by user error or side loading in the shaft. The potentiometer can be damaged. We can prevent this by using bearings for load support.

Task 3 :
After looking at the website. The product is beyond anything in the current Indian Market. For further improvements in cooking time - I think instead of changing the current design we can launch a new model with 2 pans and double the cooking capacity.

- Splitting Workflows :

In a lot of recipes, it is highly beneficial to cook in 2 different pans and then combining the results for a 1 final simmer. We can have pans mix ingredients together. For example, the sauce for a pasta being cooked in 1 pan, while the pasta being boiled in the other. Later mixed together.

- Independent Thermal Control

Each pan can have separate temperature profiles and heating control.
Example:

    Pan 1 → High heat for sautéing
    Pan 2 → Low simmer for sauce
    Enables:
        More complex recipes
        Better texture and flavor control